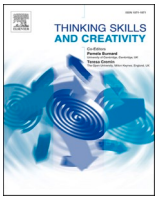
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When more or less? The role of two types of flow experiences in creative teaching

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ABSTRACT

The use of creative teaching is becoming increasingly common in higher education institutions. However, there is limited research on the impact of creative teaching on the classroom from a flow experience perspective. Using the stimulus–organism–response model and flow theory, this study presents an analysis of the overall process of creative teaching's impact on student classroom engagement, proceeding from two types of flow experiences: immediate and continuous experiences. This empirical study was conducted in the institutions piloting creative teaching in China. The results affirm the following: first, the balance of challenges and skills, goal clarity and feedback in creative teaching all affect students' immediate and continuous experiences significantly and positively. Second, immediate experiences have significantly positive effects on both behavioural and social engagement. Third, for science and engineering students, continuous experiences have significant effects on behavioural engagement and social engagement; however, these effects do not obtain for humanities and social science students. Fourth, there is no significant difference in the effects of creative teaching on engagement between male and female students. The potential contribution of this study is that we create a theoretical model of how creative teaching affects student engagement. Based on this model, educational administrators and teachers can understand the whole process of the impact of creative teaching on students' behaviour.

1. Introduction

In the training programme of cultivating talented individuals in colleges and universities, classroom teaching accounts for 70 %–80 % of the total credit hours of the training programme; hence, classroom teaching is crucial in the talent training process (Liang & Fung, 2021). Students are the main focus of the classroom, while student engagement in the classroom is an effective indicator of the quality of higher education. It is only through active participation in the activities of the university that students can improve their learning outcomes and academic performance (Avcı & Durak, 2023). Currently, the effect of classroom teaching in numerous colleges and universities worldwide is less than encouraging. Data from the China College Student Survey show that 64.4 % of students do not

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exhibit positive behavioural engagement in class. Only 33.04 % of students can ask questions or have discussions regularly in class. In addition, only 24.98 % of students can give a different point of view based on the questions asked by teachers. At present, how to encourage students to participate actively in the classroom has become a challenge for educators around the world; however, creative teaching provides viable ideas and ways to address this challenge.

Sawyer (2004) conceptualised creative teaching as a type of disciplined improvisation, highlighting the interactive and creative aspects of a teacher's role when engaging with groups of students. According to Reilly et al. (2011), teaching can be considered creative when teachers teach in a novel or unique way to develop students' understanding. Currently, many researchers confirm that creative teaching can enhance student classroom engagement. Elmaadaway (2018) studied the impact of flipped classrooms on engagement and verified that students who participated in flipped courses were more engaged than those whose classes followed traditional teaching methods. Doo et al. (2021) found that transactional distance affects learning engagement positively in large university classes using flipped learning. Yu (2023) found that a blended learning course based on the GPOB model positively impacted students' engagement and performance in learning. Ioannou (2019) proposed a gamification design model for learning using interactive tabletops that promote active student engagement. Moreover, Forman et al. (2017) confirmed that allocating a greater proportion of classroom discourse to students fosters independent thinking and encourages active engagement. Such researchers argue that the implementation of creative teaching can inspire the interest of students in learning in general and promote independent learning in particular, thus creating a student-centred classroom atmosphere and improving student classroom engagement. How creative teaching affects student engagement will be a concern for many educators, and given this, the current study proposes the following research question:

Table 1
Summary of creative teaching research.

Authors	Definition	Methodology	Main conclusions
Niu et al. (2017)	Creative teaching develops skills related to creativity through well-designed, activity-rich lessons and in-depth knowledge of mathematical domains, and stimulates students' interest in learning.	Qualitative analysis	Two unique features of Chinese creative teaching methods are as follows. On one hand, creative teaching is adaptable in addition to being novel. On the other hand, creative teaching can be achieved by learning from others.
Li et al. (2022)	Creative teaching is the use of a variety of instructional content, methods and activities by teachers to foster creativity in their students.	Quasi-experimental design method of equal group post-test design	Creative thinking teaching methods are effective in improving students' learning outcomes in multi-media material creation courses, including thinking outside the box, coming up with different ideas and answers and constantly innovating to make the outcomes presented more creative and meaningful.
Horng et al. (2005)	Creative teaching includes student-centred activities, multi-media aids, classroom management, links between the content and real life, open-ended questions and the encouragement of creative thinking.	Qualitative analysis and triangulation	Factors affecting the success of creative teaching include personality traits, family factors, learning and educational experiences, educational beliefs, educational commitment, motivation and organisational environment, with educational beliefs, educational commitment and motivation being the most important factors.
Chen and Yuan (2021)	Creative teaching means that teachers respond to the needs of their students and actively seek to innovate by designing content-rich, innovative lessons and adopting engaging pedagogical methods that enhancing students' motivation and learning outcomes effectively.	Hierarchical linear modeling	Teachers' imagination will have a positive impact on creative teaching; there is no significant relationship between headmasters' vision and teachers' creative teaching.
Huang and Lee (2015)	Creative teaching means using imaginative methods to make learning more fun, exciting and effective; aiming to develop young people's creative thinking or behaviours.	Qualitative analysis and case study	In discussing creative teaching, primary school teachers in Hong Kong agree that teachers should accommodate and enhance students' divergent thinking. Effective creative teaching must involve the active participation of students and even enable them to control their learning.
Pollard et al. (2018)	Implementation of creative teaching is unique, innovative and somewhat risky, focusing on the process of students using creative skills for learning.	Qualitative analysis	A definition of creative teaching in science, technology, engineering and maths (STEM) is summarised and the characteristics of a creative STEM teacher are reported, namely, determination, openness to the unexpected and curious, confident risk taking.
Sawyer (2015)	Creative teaching fosters creativity through specific classroom designs and teacher behaviours, where the teacher participates in the construction of knowledge with the students as facilitators and collaborators and guides them in disciplined improvisation.	Case study	Teachers are not likely to enhance students' creativity by teaching for creativity in a distinct, domain-general way. In all disciplines, the most creative learning stems from active, constructivist, collaborative and improvisational teaching strategies.
Craft et al. (2014)	Creative teaching centres on a passion for the subject and a deep understanding of the relationship between students, to foster independent thinking, using sensitive and flexible pedagogical strategies to promote egalitarian communication and thus coordinating the accumulation and construction of knowledge.	Partial mixed-methods research paradigm	The article explores the lived experience of teaching creatively in higher education from the perspectives of teachers and students. Significantly, passion for the subject is a powerful and comprehensive driver or motivator for creative teaching in a range of different subject areas and for creative teaching in higher education elsewhere.

RQ: How does creative teaching affect student engagement in the classroom?

To answer the question, this study analysed the entire process of the impact of creative teaching on student engagement using the stimulus–organism–response (SOR) model and flow theory. Because the impact of creative teaching on student classroom engagement is a continuous process, we constructed a theoretical model to analyse and explain this process. In the context of creative teaching, students are stimulated by factors such as teaching methods, assessment methods and online teaching resources and will have certain opinions and experiences about the reformed classroom, which will ultimately affect their classroom engagement.

So far, academics have focused primarily on the outcomes of creative teaching on student engagement (Cents-Boonstra et al., 2021; Chen & Chiu, 2016; Liu et al., 2017), but few studies have explored the process by which the approach influences student engagement. However, in reality, this influence process is complex as it involves the interlocking and continuous interactions of multiple factors. Therefore, this study will make up for the shortcomings of existing studies by analysing the overall influence process in detail. The theoretical contribution of this study is that we constructed a theoretical model reflecting the impact of creative teaching on student engagement, which can trace the whole process of students from being stimulated to experiencing, and then from experiencing to acting in creative teaching. The model enables higher education practitioners to understand the situation and dynamics of students at all stages of the process, thus helping them to design targeted approaches to improve student engagement. In addition, this study theoretically divides the flow experience variable in the model into two variables—immediate experiences and continuous experiences—and incorporates them into the organism stage in the SOR model, which will help this study to obtain novel and insightful findings. Furthermore, compared to the existing research, another notable aspect of this study is that it distinguishes the student groups and explores how creative teaching affects students from different disciplinary areas (science and engineering vs. humanities and social sciences) and genders (male vs. female) respectively. In this regard, the study can reveal the similarities and differences in the effects of this new teaching method across different student groups.

The remainder of this paper is organised as follows. First, we review the literature related to creative teaching, student engagement, the SOR model and flow theory. Second, we propose hypotheses and construct a research model based on the theoretical background. Third, we describe the research methodology and test the hypotheses in this study. Finally, we present the discussion, implications and limitations.

2. Theoretical background

2.1. Creative teaching

Creative teaching aims to develop the creative thinking of students and shift the focus of the classroom from teacher-delivered instruction to student learning (Wood & Ashfield, 2008). It is a concrete manifestation of teachers' creativity in the field of teaching and learning. In contrast to traditional cramming, creative teaching encourages students to think independently, participate actively and express themselves freely. Currently, the exploration of creative teaching in academia is still at the initial stage. This study summarises the relevant studies as shown in Table 1.

In this study, creative teaching is defined as adhering to the teaching concept of 'teacher-led, student-centred', optimising the teaching content and curriculum system to enhance students' creative and practical abilities. In concrete implementation, it is manifested as writing the syllabus based on outcome-based education; implementing teaching methods such as the conceive-design-implement-operate method and the project-based learning method; adopting the flipped classroom and online and offline hybrid teaching; implementing 'non-standard answer' assessment methods such as innovative essays, open projects and case studies; and producing course analysis reports for the continuous improvement of teaching content. The schools surveyed for this study specified that the method was defined as creative teaching when the teacher used any of these approaches to teach. In addition, teachers who plan to implement creative teaching for a particular course in the following semester must submit an application to the Academic Affairs Office of the university within the current semester with a detailed report on the content of the teaching. After approval by the school's teaching committee, teachers are allowed to implement creative teaching in the following semester. Furthermore, the Academic Affairs Office will assign supervisors to monitor the actual operations of these methods. Thus, the implementation of creative teaching is subject to both approval and supervision in this study, which ensures that the process of teaching reform is thereby standardised and academically rigorous.

2.2. Student engagement

Student engagement refers to the physiological and psychological energy that students invest in activities related to classroom learning, involving both intrinsic mental activity and extrinsic behaviour (Astin, 2014). Intrinsic mental activities are typically expressed through extrinsic behaviours; extrinsic behaviours are generally manifested as verbal engagement, which is the most outward, intuitive and easily evaluated form of engagement of all. The existing research has focused on how to increase the level of positive classroom engagement.

In accord with the existing literature, this study analyses the impact of the stimulus of creative teaching on individual students in terms of behavioural engagement and social engagement. Behavioural engagement is the extent to which students are impacted by the classroom climate and actively participate in classroom and school activities. (Gao et al., 2023). Social engagement refers to classroom behavioural engagement by students to satisfy real needs, motivated by external motives, i.e. value judgements and rational thinking about classroom knowledge.

2.3. SOR model

The SOR model suggests that external stimuli can influence an individual's internal state and, thus, their behavioural responses (Mehrabian & Russell, 1974). The model comprises three parts: stimulus, organism and behavioural responses.The S (stimulus) denotes the exterior force that is capable of influencing an individual's mental state, O (organism) represents the interior state, which includes affective and cognitive behaviours, and R (response) marks the final behavioural outcome of an individual. Pandita et al. (2021) considered the SOR model to be the classic paradigm of explaining how exogenous incentives interplay with an individual's intrinsic state to thereby lead to the individual's behaviour.

In this study, we explored the mechanism of the influence of creative teaching on student classroom engagement (R) using the stimuli that students were exposed to under influence of the creative teaching as the antecedent variable (S) and their immediate and continuous experiences as the mediating variables (O). In terms of the stimulation, we combined the SOR model with flow theory and chose the three dimensions of challenge and skill balance, goal clarity and feedback to measure the motivation students receive. The balance of challenge and skill occurs when an individual's skills are balanced with the challenges that they undertake. When people perceive that challenge and skill are balanced and are at a high level, the flow experience emerges (Csikszentmihalyi, 1988). Goal clarity not only facilitates students' concentration but also enables them to grasp the learning process more accurately, making it easier for them to enter a state of flow. Feedback is a summary and reflection of learning outcomes. Feedback is a summary and reflection of learning outcomes. Feedback on successes and failures affects students' retrospective evaluations of performance and hints at probabilistic information for future success or failure (Pekrun, 2006). In addition, feedback must be rationally related to goal clarity to facilitate the generation of flow. Regarding the organism aspect, we divide the flow experience generated by students into continuous and immediate experiences. Finally, for the response, we chose students' behavioural engagement and social engagement as the variables for analysis.

2.4. Flow theory

The notion of flow refers to the positive mental states that people exhibit while focused on performing a certain behaviour or immersed in a certain situation, which usually occurs when an individual's perceived challenge is balanced with the ability to meet the challenge and complete the task (Beard, 2015). When individuals are in flow, they are completely absorbed in what they are engaged in, filtering out irrelevant perceptions, forgetting about time and their external environment and deriving satisfaction and pleasure simply from the process of the activity (Csikszentmihalyi, 1988). The psychological concept of flow is gradually being applied to other research areas such as consumer behaviour, human-computer interaction, education and training (Rodríguez-Torrico et al., 2023; Van Schaik et al., 2012; Yoo et al., 2018). In particular, according to existing research results, flow exists within the classroom, and the flow experience has a significant positive impact on learning outcomes (Egbert, 2004; Mueller & Wulf, 2022; Shin, 2006).

In the context of creative teaching, interesting classroom activities, personalised learning experiences, positive feedback and support and challenging tasks stimulate students' interest and motivation to learn. This enjoyable learning atmosphere allows them to be fully engaged in learning and enjoy the learning process, resulting in a flow experience. In this study, to explore in depth the flow experience, we classified it as immediate experience and continuous experience and set different variables to conduct specific analyses. Immediate experiences are real-time flow states that arise over a short period after students are stimulated by flow antecedents. We chose concentration, the merging of activity and awareness and perceived control for a specific analysis. Concentration is the key to flow experience and refers to a high level of focus by students in the current classroom. The merging of activity and awareness refers to a state of total concentration where a student's thoughts completely dominate their behaviour. Perceived control refers to a student's better sense of control over the current activity and their belief that they can deal with the upcoming subsequent behaviour. Meanwhile, the concept of continuous experiences denotes a state of flow that can continue over a longer period, and we specifically chose the transcendence of self, the transformation of time and autotelic experience for analysis. The transcendence of self is when students are concentrating so hard on the current task that they lose sight of themselves; the transformation of time is when a student experiences a temporary distortion in the perception of time (e.g. the feeling that time passes faster than usual) and the autotelic experience is a temporary loss of self-awareness (e.g. forgetting one's own identity or forgetting that you are hungry or tired.

2.5. Integration of the SOR model with flow theory

In this study, we theoretically integrated two types of flow experience (immediate experiences and continuous experiences) into the organism stage of the SOR model and measured the organismal state based on different flow dimensions. The main reasons for combining the above two are as follows. First, the two theories have common elements that can be integrated. Specifically, flow experience emphasises that people enter a state of immersion when they are fully engaged in a situation, focusing attentively and having filtered out all irrelevant perceptions while performing an activity (Csikszentmihalyi, 1990). The SOR model highlights the existence of an organism stage between stimulus and response. It includes the internal physiological and psychological factors of the individual (Yang et al., 2021). Both theories focus on the individual's psychological response and state in a particular situation, and centre on the individual's processing of stimuli, which provides a unified view of how internal states affect subsequent responses. Second, there are similarities between the organism stage and the flow experience. The organism stage represents how an individual's internal physiological and psychological factors affect the final response after receiving an external stimulus. This stage involves psychological processes such as the individual's cognitive processing and emotional response to the stimulus. Similarly, the flow state is also a highly engaged and focused psychological state achieved by the individual during a particular activity, which filters and

processes external stimuli more effectively. Based on this, the flow experience can be regarded as a special manifestation of the organism stage in the SOR model, i.e. when the organism is in the flow state, its processing and response to the stimulus is more efficient and deeper. By integrating flow theory with the SOR model, this research can better understand how the psychological processes involved in flow enhance the effectiveness of responses to external stimuli. This integration highlights how a state of flow can lead to more efficient processing and deeper engagement, thereby enriching the overall explanatory power of the SOR model. Therefore, integrating the flow theory with the organism stage of the SOR model helps us to reach a deeper understanding of an individual's psychological responses and states in a given situation, and thereby provides useful guidance for practical applications (e.g. instructional design or learning strategies).

3. Research hypotheses

3.1. Impact of the stimulus stage on the organism stage

3.1.1. Balance of challenge and skill

Challenge entails the possibility and opportunity for an individual to grow, and skill is their ability to deal with the challenge (Bricteux et al., 2017). The balance of challenge and skill is defined as the level of a perceived challenge of an individual's activity being within the range of what their skills can control, which is an important predictor of a flow state (Fong et al., 2015; Voelkl & Ellis, 1998). Csikszentmihalyi (1988) highlighted that if a challenge is too difficult or beyond an individual's ability, then it may lead to anxiety, stress and frustration, inhibiting the production of a flow experience; if a challenge is too easy to accomplish without much skill or effort, then it may lead to boredom and a loss of interest, making it similarly difficult to enter the state of flow. The implementation of creative teaching has meant that students will be more involved in hands-on learning and teamwork and will face higher learning standards and more rigorous assessment requirements. Students will be more challenged and competitive due to the changes in learning styles and the attendant need for interdisciplinary learning. When students' skills are balanced with the difficulty of these challenges and they address the problems successfully, they will feel confident and satisfied, and this self-efficacy helps motivate students and facilitates the flow experience. We believe that the balance of challenge and skill will have a notably positive impact on students' flow experience and thus propose the following hypotheses:

H1a: The balance of challenge and skill has a notably positive impact on immediate experiences.

H1b: The balance of challenge and skill has a notably positive impact on continuous experiences.

3.1.2. Goal clarity

Goal clarity indicates that students know exactly what they want to gain from classroom learning, how to do it and how far they want to go. Individuals tend to maintain a high level of concentration when they have a clear goal to guide them in their tasks (Quinn, 2005). Goal clarity leads to higher levels of task performance than vague abstract goals (Locke & Latham, 2006), and self-efficacy is a potential mediator of a goal–performance relationship (Schmidt, 2019). Creative teaching is the process of using innovative teaching methods to make the objectives of the course understandable to students and establishing the visualisation steps to that end. Self-determination theory links the pursuit of flow theory to intrinsic motivation (Moneta & Csikszentmihalyi, 1996). The clearer the student's goals, the higher the level of personal motivation, and thus the more disciplined the behaviour, and the higher grades (Meng et al., 2023). When a student achieves a goal, the self-efficacy gained will motivate them to move closer to the next goal, resulting in a flow experience. Therefore, we believe that goal clarity will have a notably positive impact on individual students' flow experience and propose the following hypotheses:

H2a: Goal clarity has a notably positive impact on immediate experiences.

H2b: Goal clarity has a notably positive impact on continuous experiences.

3.1.3. Feedback

Feedback means not only that students' questions are answered promptly but also that students can be informed by teachers, students, etc., whether their performance is acknowledged. Individuals are more likely to have a flow experience when immediate and explicit feedback enables them to focus and concentrate on the current activity, to perceive the current task as enjoyable and within their ability to control and thereby feel confident about the outcome of the task (Stavrou et al., 2015). According to reinforcement theory, we can regard flow as positive feedback that reinforces an individual's behaviour (Teng et al., 2012). Creative teaching emphasises the implementation of assessment throughout the whole process of teaching and learning following the course content and educational needs. When students complete the process tasks, they will receive positive encouragement from the teacher and various sensory stimuli, thus obtaining rich and positive emotional experiences. These emotional experiences will have a motivating impact on students, i.e. the positive reinforcement of students' behaviour, which will make them feel satisfied and encourage them to remain engaged in the classroom activities and thus obtain a flow experience. Therefore, we believe that feedback will have a notably positive impact on students' mental flow experience and propose the following hypotheses:

H3a: Feedback has a notably positive impact on immediate experiences.

H3b: Feedback has a notably positive impact on continuous experiences.

3.2. Impact of the organism stage on the response stage

3.2.1. Immediate experiences

Immediate experiences refer to a particular moment when an individual feels that they are in a state of complete dedication, concentration and forgetfulness. Creative teaching inspires students' interest in learning and increases their enthusiasm and engagement through the adoption of interactive pedagogy, the introduction of novel curricula and the provision of diversified training resources. In an educational setting, a total commitment to classroom activities promotes an optimal learning experience for students (Shernoff et al., 2003). When students reach the state of flow, they eliminate all extraneous emotions and thoughts, focus on the current activity, feel fulfilled and joyful (Mirvis, 1991), and reach a state where their thoughts completely dominate their behaviour and believe that they are capable of meeting the challenge and solving the problem successfully. This kind of interest, motivation and positivity will prompt them to ask questions and share their opinions. Therefore, this study concluded that immediate experiences have a predictive influence on enhancing students' behavioural engagement. Patrick et al. (2007) argued that the social and emotional environment in the classroom is a prerequisite for student engagement in activities and tasks. When students have immediate experiences, they believe that they are in control and successful, and this positive emotional environment promotes students' engagement in classroom communication and collaboration with others. Therefore, we conclude that immediate experiences have predictive influences on enhancing students' social engagement. Given this, we propose the following hypotheses:

H4a: Immediate experiences have a notably positive impact on behaviour engagement.

H5a: Immediate experiences have a notably positive impact on social engagement.

3.2.2. Continuous experiences

Continuous experiences are those in which an individual experiences multiple consecutive immediate experiences over some time. Creative teaching provides an enjoyable learning environment for students to give their full attention to classroom activities (Horng et al., 2005) which makes it easier for them to enter the state of flow. Students in the flow forget about time and surrounding distractions, temporarily lose self-awareness (Chen, 2000), have motivation and enthusiasm and are continuously engaged in the activity. This sustained engagement helps students to be more focused on a given task and improves their learning efficiency, thus enhancing behavioural engagement. In addition, creative teaching increases motivation and improves the learning atmosphere, which leads to a highly intense sense of belonging, and consequently to enjoyment and stimulation (Miranda et al., 2023). Students' continuing flow experience in the classroom allows them to become more comfortable with the classroom environment and establishes a sense of identification with the session, which leads to increased motivation for social interaction and the demonstration of higher levels of social engagement. Therefore, we develop the following hypotheses:

H4b: Continuous experiences have a notably positive impact on behaviour engagement.

H5b: Continuous experiences have a notably positive impact on social engagement.

Fig. 1 depicts the theoretical model for this study.

4. Methodology

In September 2022, Shandong Province decided that creative teaching would be implemented in colleges and universities across the province within 5 years to improve the quality of training of talented undergraduates. Because creative teaching involves a variety of concepts, methods and assessment methods, the provincial Department of Education planned to spend 3 years exploring and

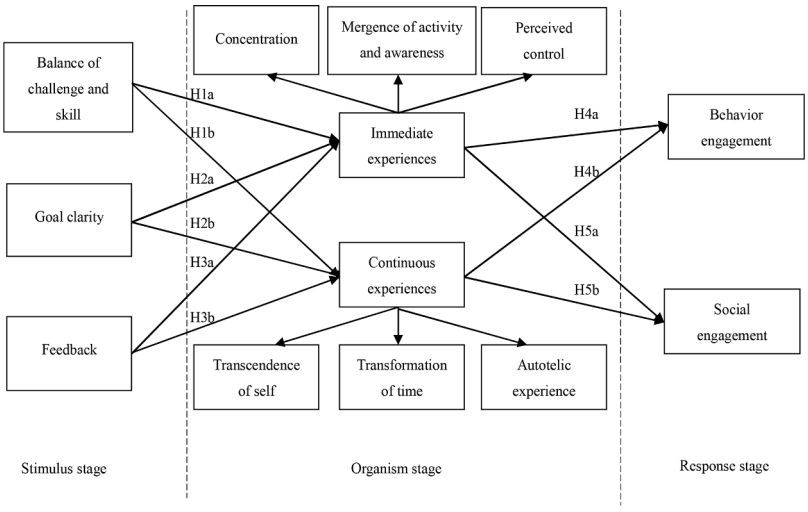


Fig. 1. Theoretical model.

examining this method. The institution commissioned by this study is the first batch of pilot institutions of creative teaching in the province and has carried out creative teaching for more than a year. The implementation of creative teaching in this university is mainly reflected in the following aspects. First, in terms of educational philosophy, teachers adhere to a student-oriented teaching perspective and learning process. Secondly, in terms of the content of teaching, teachers are encouraged to carry out personalised teaching according to the characteristics of students. Third, in the teaching mode, teachers strengthen the integration of information technology and classroom teaching. Fourth, in terms of teaching methods, teachers transform students from 'bystanders' to 'actors' in class using heuristic teaching, interactive teaching, exploratory discussions, etc. Fifth, in the aspect of academic evaluation, teachers focus on process evaluation and emphasise ability evaluation. Finally, in terms of teaching evaluation, teachers are required to formulate a scientific and rigorous evaluation programme of the whole process, and constantly improve the implementation of creative teaching.

4.1. Measures

A Likert scale was used to obtain data for this study. Based on the theoretical model, we designed 41 survey items for the 11 variables. Each item was scored on a 7-point scale, with 1 indicating strong disagreement and 7 indicating strong agreement. For each

Table 2
Specific items of this scale.

Variables	Items	References
Balance of challenge and skill (BCS)	BCS1: In the classroom, I was challenged, but I believed that my skills would allow me to meet the challenge. BCS2: In the classroom, my abilities matched the high challenge of the situation. BCS3: In the classroom, I felt that I was competent enough to meet the high demands of the situation. BCS4: In the classroom, the challenge and my skills were at an equally high level.	Jackson and Marsh (1996)
Goal clarity (GC)	GC1: During the class, I knew clearly what I wanted to do. GC2: During the class, I had a strong sense of what I wanted to do GC3: During the class, I knew what I wanted to achieve. GC4: During the class, my goals were clearly defined.	Jackson and Marsh (1996); Guo and Poole (2009)
Feedback (FB)	FB1: During the class, it was really clear to me that I was doing well. FB2: During the class, I was aware of how well I was performing. FB3: During the class, I had a good idea about how well I was doing. FB4: During the class, I could tell by the way I was performing how well I was doing.	Jackson and Marsh (1996); Guo and Poole (2009)
Merging of activity and awareness (MAA)	MAA1: I made the correct movements in class without thinking about trying to do so. MAA2: Things just seemed to be happening automatically during the class. MAA3: I performed automatically in the class. MAA4: I did things spontaneously and automatically without having to think in the class.	Jackson and Marsh (1996)
Perceived control (PC)	PC1: During the class, I feel in control. PC2: I feel that I have no control over my interaction with teacher in the class. PC3: The classroom allows me to control my interaction with teacher.	Agarwal and Karahanna (2000)
Concentration (CO)	CO1: My attention was focused entirely on what I was doing in the class. CO2: It was no effort to keep my mind on what was happening in the class. CO3: I had total concentration in the class. CO4: I was completely focused on the task at hand during the class.	Jackson and Marsh (1996)
Transformation of time (TT)	TT1: Time appeared to go by very quickly when I am in the class. TT2: I lost track of time during the class. TT3: Time flew in the class.	Guo and Poole (2009)
Transcendence of self (TS)	TS1: I was not concerned with what others may have been thinking of me in the class. TS2: I was not concerned with how I was presenting myself in the class. TS3: I was not worried about what others may have been thinking of me in the class.	Guo and Poole (2009)
Autotelic experience (AE)	AE1: I really enjoyed the experience during the class. AE2: I loved the feeling experienced in the class and want to capture it again. AE3: The experience in the class left me feeling great. AE4: I found the experience extremely rewarding during the class.	Guo and Poole (2009)
Behavioural engagement (BE)	BE1: I take notes when I participate in classroom discussions. BE2: I stay focused during classroom activities. BE3: I talk about classroom topics even when I am not in it. BE4: I complete all classroom tasks on time.	Hoi and Le Hang (2021)
Social engagement (SE)	SE1: I share learning materials with other classmates. SE2: I build on other ideas during classroom discussions. SE3: I ask teachers if I do not understand something when learning in the classroom. SE4: I respond to other classmates' questions after the class.	Hoi and Le Hang (2021)

item, we drew on high-quality studies previously published in SSCI journals in our design. The survey items used in the study are shown in Table 2.

4.2. Data collection

For this study, we commissioned the Academic Affairs Office of a university to distribute the questionnaires. First, we implemented a pre-survey in which 60 questionnaires were distributed to 12 faculty departments to test the suitability of the questionnaires for this research. The results confirmed that all the survey questions were focused on creative teaching and were clear and easy to understand.

For the formal survey, we used the Questionnaire Star website to implement the survey. QuestionStar.com is a specialised online platform in China that offers users functional and personalised services, such as online questionnaires. We designed the questionnaire on the Questionnaire Star website and distributed the URL of the questionnaire along with the study's objectives, the survey participation method and the consent form to each student's email address through the faculty system. The survey was voluntary, and it was up to the students to choose whether or not to participate in the survey. There were no repercussions concerning whether they participated or not, but to ensure a high participation rate, we encouraged students to participate in the survey through email. Considering that not all courses have adopted creative teaching methods over the past year and that students are allowed to choose freely the courses to attend, we included the following screening question in the survey: Have you ever experienced creative teaching methods? Only those students who chose 'yes' were considered valid samples.

The period of this data collection was from 8 to 19 January 2024, which coincides with the last 2 weeks of the current semester. During this period, all course content for the semester had been taught. Some course exams had been completed and exams for some courses were about to begin. Therefore, students were able to respond thoughtfully to the questionnaire from the perspective of the overall implementation of the course after a semester-long creative teaching experience, thus ensuring that the data were comprehensive and accurate. In total, we distributed 33,345 questionnaires and received 9847 responses. Of these, 4864 students experienced creative teaching methods. Subsequently, we filtered the questionnaires based on whether the responses were repeated over a wide range. Finally, 4454 valid questionnaires were obtained. The attribute information of valid samples is shown in Table 3.

4.3. Data analysis

In this study, Lisrel 8.7 software and the structural equation modelling (SEM) method were used to test the research hypotheses in the theoretical model and to evaluate the statistical significance of the path coefficients in it. SEM is a kind of statistical analysis method to describe the complex direct and indirect interaction between variables, which has been widely used in social science studies. Structural equation modelling is divided into two categories, covariance-based SEM (CB-SEM) and partial least squares SEM (PLS-SEM). Covariance-based SEM is more suitable than PLS-SEM for the confirmation of the theoretical model. Therefore, CB-SEM is used to test the theoretical model and each hypothesis in the model. In this study, the main steps of CB-SEM are as follows. First, we analysed the descriptive statistics of the measurement items. Second, we analysed the reliability and validity of the scale. Third, we tested the theoretical models and research hypotheses in detail.

5. Results

5.1. Descriptive statistics

Table 4 presents the descriptive statistics of all items. Respondents were generally positive about all items with a mean of 5.21, with BE (5.45) having the highest mean and TS (4.64) having the lowest. The Kolmogorov–Smirnov test shows that each item is significantly different from the standard normal distribution. In addition, the results of the non-parametric test show that there is no significant difference in the mean and standard deviation of all the items measuring a construct. According to the research by Wijaya et al. (2022b), the limit for skewness and kurtosis is |2.3|. Therefore, both kurtosis and skewness in this study were acceptable.

5.2. Reliability and validity test

Before conducting a formal statistical analysis, we tested the reliability and validity of the scales to ensure that they were

Table 3
Attribute information of the respondents.

Demographics	Category	Frequency	Percentage (%)
Gender	Male	2084	46.79
	Female	2370	53.21
Grade	Freshman	869	19.51
	Sophomore	1661	37.29
	Junior	1491	33.48
	Senior	433	9.72
Speciality type	Science and engineering	3011	67.60
	Humanities and social sciences	1443	32.40

Table 4
Descriptive statistics of all measurement items.

Variables	Items	Mean	Standard deviation	Skewness	Kurtosis
BCS	BCS1	5.40	1.18	-0.90	1.15
	BCS2	5.34	1.22	-0.84	0.91
	BCS3	5.35	1.21	-0.88	1.11
	BCS4	5.32	1.22	-0.80	0.77
GC	GC1	5.39	1.17	-0.86	1.15
	GC2	5.41	1.17	-0.86	1.09
	GC3	5.39	1.16	-0.86	1.15
	GC4	5.37	1.17	-0.83	1.02
FB	FB1	5.30	1.19	-0.74	0.72
	FB2	5.45	1.10	-0.80	1.10
	FB3	5.44	1.12	-0.83	1.07
	FB4	5.37	1.15	-0.76	0.87
MAA	MAA1	5.10	1.29	-0.57	0.07
	MAA2	5.38	1.15	-0.80	0.99
	MAA3	5.27	1.22	-0.71	0.51
	MAA4	4.98	1.35	-0.52	-0.07
PC	PC1	5.05	1.30	-0.54	0.06
	PC2	4.67	1.44	-0.38	-0.35
	PC3	5.14	1.21	-0.53	0.28
CO	CO1	5.18	1.20	-0.61	0.41
	CO2	4.99	1.29	-0.49	-0.02
	CO3	5.19	1.17	-0.53	0.30
	CO4	5.16	1.21	-0.63	0.50
TT	TT1	5.14	1.22	-0.60	0.41
	TT2	5.10	1.24	-0.57	0.31
	TT3	5.07	1.28	-0.58	0.31
TS	TS1	4.88	1.38	-0.51	-0.21
	TS2	4.64	1.45	-0.37	-0.45
	TS3	4.83	1.37	-0.46	-0.26
AE	AE1	5.25	1.15	-0.63	0.70
	AE2	5.20	1.18	-0.62	0.57
	AE3	5.23	1.16	-0.62	0.62
	AE4	5.28	1.13	-0.69	0.93
BE	BE1	5.29	1.15	-0.69	0.82
	BE2	5.32	1.11	-0.67	0.85
	BE3	5.22	1.19	-0.68	0.66
	BE4	5.45	1.08	-0.71	0.93
SE	SE1	5.48	1.06	-0.76	1.21
	SE2	4.97	1.28	-0.51	0.15
	SE3	5.30	1.10	-0.64	0.99
	SE4	5.36	1.13	-0.85	1.46

Note: BCS = balance of challenge and skill; GC = goal clarity; FB = feedback; MAA = mergence of activity and awareness; PC = perceived control; CO = concentration; TT = transformation of time; TS = transcendence of Self; AE = autotelic experience; BE = behavioural engagement; SE = social engagement.

scientifically sound and reasonable. We used two software programmes, SPSS 23.0 and Lisrel 8.7, to perform a series of analyses. First, we conducted an exploratory factor analysis (EFA) to understand the affiliation between variables and question items. The EFA results revealed a Kaiser–Meyer–Olkin test score of 0.885, demonstrating that the data acquired were appropriate for factorial analysis (Greenier et al., 2023). However, the factor loadings of the items FB4, MAA2 and SE2 were all <0.7, suggesting that the correlation between these items and the corresponding variables was weak, therefore, we deleted these three items (Dash and Paul, 2021). Except for these items, all other items were involved in the reliability and validity tests of the scale. Table 5 shows the factor loading of all other items.

In the reliability test, the results in Table 5 confirm that the composite reliability and Cronbach's alpha values for the six variables exceeded 0.7, demonstrating that the validity of the scale is acceptable (Khan & Abbas, 2022). In the validity analysis, the average variance extracted (AVE) values of all variables exceeded 0.5, indicating that the scale has good convergent validity (Elnadi & Gheith, 2023). In addition, the Fornell-Larcker test of Table 6 shows that the open-square value of the AVE value for each variable exceeds the correlation between that variable and the other variables, which indicates that the scope has good differentiation validity (Kupczyszyn et al., 2024). Furthermore, the heterotrait-monotrait ratio of correlations value does not exceed 0.9 in Table 6, proving that the discriminant validity between the latent variables is acceptable (Wijaya et al., 2022a). Overall, the scale designed in this study has good reliability and effectiveness.

5.3. Hypothesis tests

We used SEM to develop the statistical analysis, and the results show that the theoretical model fits the empirical data well,

Table 5
Statistical results of some indicators.

Variables	Items	Loadings	AVE	CR	Cronbach alpha
BCS	BCS1	0.77	0.62	0.87	0.95
	BCS2	0.79			
	BCS3	0.80			
	BCS4	0.78			
GC	GC1	0.80	0.64	0.87	0.95
	GC2	0.80			
	GC3	0.80			
	GC4	0.79			
FB	FB1	0.77	0.56	0.79	0.91
	FB2	0.74			
	FB3	0.73			
MAA	MAA1	0.77	0.55	0.78	0.88
	MAA3	0.73			
	MAA4	0.72			
PC	PC1	0.78	0.58	0.80	0.90
	PC2	0.75			
	PC3	0.75			
CO	CO1	0.71	0.55	0.83	0.90
	CO2	0.79			
	CO3	0.71			
	CO4	0.75			
TT	TT1	0.75	0.56	0.79	0.93
	TT2	0.76			
	TT3	0.73			
TS	TS1	0.78	0.67	0.86	0.95
	TS2	0.85			
	TS3	0.83			
AE	AE1	0.79	0.65	0.88	0.95
	AE2	0.79			
	AE3	0.81			
	AE4	0.83			
BE	BE1	0.74	0.56	0.84	0.91
	BE2	0.77			
	BE3	0.71			
	BE4	0.77			
SE	SE1	0.76	0.55	0.78	0.88
	SE3	0.74			
	SE4	0.72			

Note: BCS = balance of challenge and skill; GC = goal clarity; FB = feedback; MAA = mergence of activity and awareness; PC = perceived control; CO = concentration; TT = transformation of time; TS = transcendence of Self; AE = autotelic experience; BE = behavioural engagement; SE = social engagement.

Table 6
Statistical results of discriminant validity.

Factors	BCS	GC	FB	MAA	PC	CO	TT	TS	AE	BE	SE
BCS	0.79										
GC	0.72	0.80									
FB	0.67	0.73	0.75								
MAA	0.62	0.64	0.68	0.74							
PC	0.58	0.58	0.61	0.69	0.76						
CO	0.56	0.58	0.60	0.60	0.64	0.74					
TT	0.51	0.51	0.53	0.57	0.60	0.66	0.75				
TS	0.26	0.24	0.27	0.39	0.42	0.42	0.44	0.82			
AE	0.53	0.54	0.55	0.56	0.58	0.63	0.65	0.41	0.81		
BE	0.50	0.54	0.55	0.54	0.54	0.61	0.59	0.36	0.72	0.75	
SE	0.47	0.51	0.53	0.49	0.49	0.54	0.54	0.31	0.64	0.72	0.74

Note: BCS = balance of challenge and skill; GC = goal clarity; FB = feedback; MAA = mergence of activity and awareness; PC = perceived control; CO = concentration; TT = transformation of time; TS = transcendence of Self; AE = autotelic experience; BE = behavioural engagement; SE = social engagement.

indicating that the hypothesis testing results obtained based on the model are scientifically valid (Shi et al., 2023). Fig. 2 shows the analytical results of hypothesis testing and shows that the probability p-value of the path coefficients of the 10 research hypotheses is <0.001, which indicates that the causal relationships between the variables are all significant (Sivaci and ALTAŞ, 2023). Thus, all the research hypotheses in this study are supported by the empirical data.

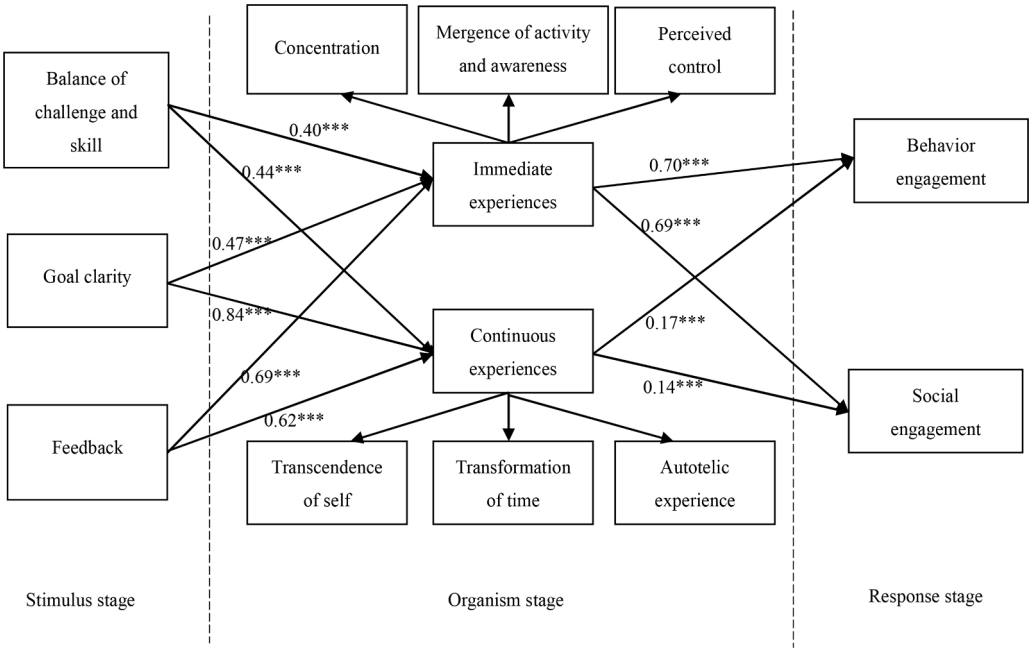


Fig. 2. Full model hypothesis testing.

According to Finnegan et al. (2008), there are differences in the behaviour of students in different majors; hence, their levels of engagement vary. Therefore, to gain further insights, we analysed the similarities and differences between students majoring in science and engineering and those majoring in humanities and social sciences throughout the engagement process. Fig. 3 illustrates the engagement process of students majoring in science and engineering; and Fig. 4 shows the engagement process of those majoring in humanities and social sciences. By comparing the results of these two hypothesis tests, we can see some variability in the process of student engagement between different majors. Specifically, among science and engineering students, continuous experiences have a marked effect on behaviour engagement and social engagement, i.e. H5a and H5b are supported; however, for humanities and social science students, this significant impact is absent, i.e. H5a and H5b are not supported. Thus, there is variability in the impact of creative teaching on different categories of students.

According to Santos et al. (2021), there are also differences in the level of student engagement by gender. Therefore, to ascertain the influence of gender on creative teaching, we analysed the variability of the learning engagement process between boys and girls. The results showed that all the research hypotheses were supported for boys and girls. Therefore, there is no significant difference in the impact of creative teaching on students of different genders.

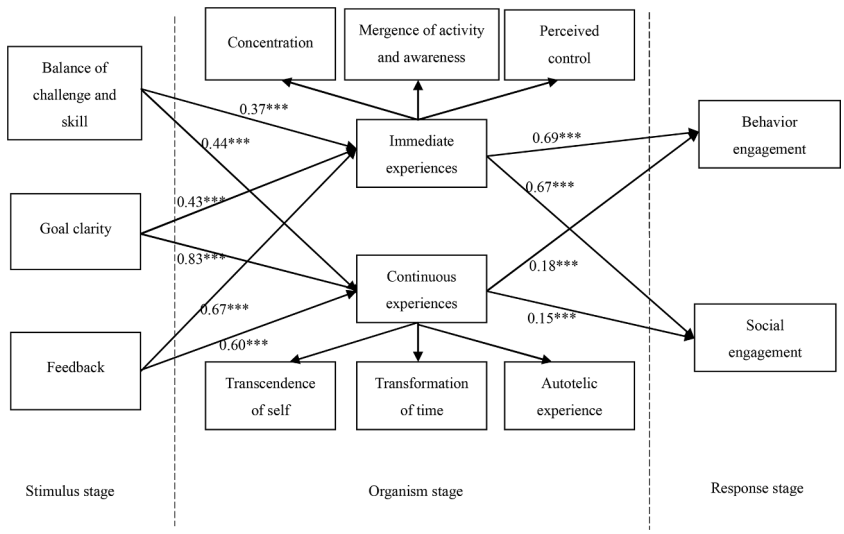


Fig. 3. Hypothesis testing for science and engineering students.

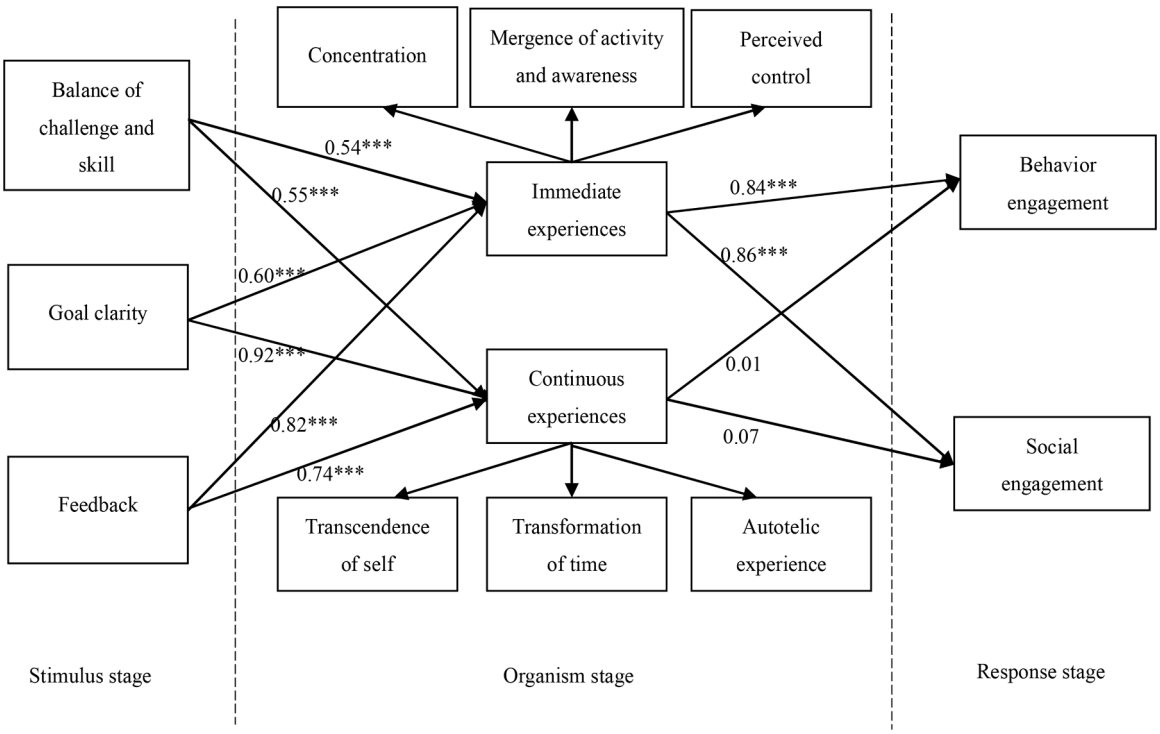


Fig. 4. Hypothesis testing for humanities students.

6. Discussion

Using the SOR model and flow theory, in this study, we explored the process by which creative teaching affects college students' classroom engagement and drew four novel and valuable conclusions.

First, in creative teaching, the balance of challenge and skill, goal clarity and feedback all have a significant positive impact on students' immediate and continuous experiences. This is consistent with previous findings in traditional instructional settings (Egbert, 2004; Mandhana & Caruso, 2023). Creative teaching engenders a positive learning atmosphere in which students can achieve a balance of challenge and purpose, set clear goals and receive timely feedback that is pleasurable and fulfilling, thus facilitating flow experience. Such experiences brought about by creative teaching can be either short-lived or long-lasting. Specifically, it may manifest itself as a short-lived experience of flow within a particular phase of a student's learning, or it may be transformed into a stable, enduring source of motivation for the student's long-term learning process.

Second, immediate experiences affect behaviour engagement and social engagement significantly and positively. This finding is consistent with those of previous studies (Mesurado et al., 2016). Creative teaching emphasises the active participation of students, especially encouraging them to engage in classroom discussions and group work. This enables students to exchange learning outcomes instantly in practice and deepen their understanding of knowledge through interaction and collaboration. In this process, students experience a realisation of their self-worth, forget about anxiety and low self-esteem and obtain immediate experience, which prompts them to participate more actively in classroom activities and establish a positive interactive relationship with teachers and other students.

Third, for science and engineering students, continuous experiences can have a marked influence on behaviour engagement and social engagement; however, these effects do not happen for humanities and social science students. This finding adds to the existing research (Chung & Pan, 2023; Hamari et, al.,2016). We believe that the main reasons for it are as follows. Science and engineering students must usually apply various skills to solve specific problems, and this kind of thinking ability and learning style makes it easier for them to enter the flow state. In this state, these students can improve their skills continuously in practice, increase the efficiency of their study and work and gain a sense of achievement, which subsequently increases their level of engagement. The study and work of humanities and social sciences students may involve more activities, such as pondering, reading and writing and less specific skill applications than those of science and engineering students. Therefore, flow theory may be less significant for them.

Finally, there is no significant difference in the effect of creative teaching on the engagement of boys and girls. This is different from previous findings on the effect of creative teaching on boys and girls (Khan et al., 2017; Lin et al., 2020). Previous research concluded that girls outperform boys in creative teaching situations in terms of engagement and learning outcomes. One possible explanation for this is the influence of cultural and social factors on participants. Due to gender expectations and societal pressures, girls are more likely inclined to seek parental and teacher approval by excelling, whereas this motivation may not be prevalent among boys. In the creative teaching context of this study, modern teaching methods focus more on individual differences than simply the learning

differences between genders. Changes in the social environment and increased awareness of gender equality will lead to boys and girls being treated more equally and fairly in the classroom and thereby being more balanced in their participation. In this sense then, this finding does not show significant differences between genders.

7. Implications

7.1. Theoretical implications

This study makes some theoretical and practical contributions. Theoretically, we develop a strategic model of how creative teaching affects student engagement. In one aspect, we applied the SOR model from the field of psychology to the field of education, thereby extending the application area of the model and highlighting its versatility at the level of teaching effectiveness. Based on this theoretical model, educational administrators and teachers can better understand the whole process of the impact of creative teaching on students' behaviour. This process goes from the stimulation of pedagogical elements to the students' psychological responses and from the students' psychological responses to their actual engagement. The model was supported by the empirical results. In another aspect, we theoretically divided the flow experiences into immediate and continuous experiences and integrated these two types of flow experiences with the organism stage of the SOR model to explore the impact of creative teaching on student engagement from both short-term and long-term perspectives, disclosing the similarities and differences between the two types of flow experiences that play a role in creative teaching. The results of the empirical analyses both support the relevant findings in traditional teaching and reveal the unique characteristics of creative teaching.

7.2. Practical implications

From a practical standpoint, there are two ways that educators can begin to maximise student engagement. First, for science and engineering students, the emphasis is on activating their immediate and on-going experiences to promote engagement. Science and engineering students tend to deepen their understanding of theoretical knowledge through practical activities. Engaging in practical experiences allows these students to better understand and internalize theoretical concepts by observing their real-world applications. Hence, teachers should design practical application scenarios related to the course content, so that students can experience first-hand the process of transforming theoretical knowledge into practical application. For example, lab experiments, engineering projects, or fieldwork can be integrated into the curriculum. This not only enhances their interest in learning significantly but also provides them with a richer immediate and continuous experience in learning, promoting their learning engagement effectively. This hands-on approach effectively bridges the gap between theory and practice, promoting sustained engagement and a richer learning experience. Furthermore, for students in the humanities and social sciences, the focus is on activating their immediate experiences to promote engagement. Humanities and social sciences students are required to think critically. Therefore, teachers can arrange for students to participate in discussions and debates so that they can express their views and ideas instantly in these interactive platforms and practice their logical thinking skills, from which they can gain immediate flow experience and feedback. Immediate flow experiences can significantly enhance students' engagement with course material. By incorporating interactive elements, teachers can create a more dynamic and responsive learning environment that fosters psychological experiences and stimulates actual participation.

8. Conclusions and limitations

This study combined the SOR model and flow theory to analyse the entire influencing process of creative teaching on students' classroom engagement and drew four valuable conclusions. First, balancing challenge and skill, goal clarity and feedback in creative teaching all have a significant positive effect on students' immediate and continuous experience. Second, immediate experience has a significant positive effect on behavioural and social engagement. Third, for science and engineering students, continuous experience has a significant effect on behavioural and social engagement; however, these effects do not appear in the humanities and social science fields. Fourth, there is no significant difference in the effects of creative teaching on engagement between male and female students.

There are five limitations to this research. First, it relies primarily on cross-sectional survey data and lacks longitudinal tracking data, which makes it difficult for the study to determine how the variables in the theoretical model evolve or influence each other over a long period of time. To better determine the temporal relationship between variables, future research should include time series data and long-term observations of students. Additionally, incorporating methods beyond questionnaires, such as teacher–student interviews and classroom observations, would enhance the robustness of the data. Second, this study was conducted within the current teaching reform context. With the continuous advances in teaching philosophies and methods, the connotation and extension of creative teaching may expand. Then, their impact on students' behaviours may be more complex. Therefore, future research should consider examining this issue within the latest creative teaching context. Third, flow theory and the SOR model each focus on specific variables, but when integrating the two theories, some equally important variables may be overlooked, which could also play a role in certain creative teaching environments. Therefore, future studies could seek other integrating elements to compensate for this limitation. Fourth, considering that colleges and universities in Shandong province are currently at the exploration stage of creative teaching, we do not divide and explore the depth of flow experience in our research. Moreover, phenomenological data analysis is not yet mature. As creative teaching matures, we expect to be able to combine triangulation with phenomenological data analysis methods. This combination will enable us to explore the flow experience in greater depth and provide more comprehensive and

detailed conclusions about how creative teaching methods affect student engagement and learning outcomes. Fifth, this study relies solely on quantitative analysis methods and does not incorporate qualitative analysis. The lack of qualitative data limits a deeper understanding of students' flow experiences. Future research should integrate qualitative methods, such as thematic analysis or grounded theory to provide richer and more nuanced insights.

Author statement

For the completion of this research work, we can declare that in all the work of this paper, we did not use generative artificial intelligence technology.

With respect to the provenance of the work, we can declare that this paper presents original, previously unpublished work. It is not currently under consideration elsewhere and has not been submitted before to this or any other journal. There are no conflicts of interest to declare.

Sincerely, and on behalf of the authors.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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Wenlong Zhu: Conceptualization, Funding acquisition, Project administration, Software, Supervision, Validation, Writing – original draft, Writing – review & editing. **Xinmiao Wang:** Conceptualization, Data curation, Methodology, Investigation, Formal analysis, Writing – original draft. **Yong Yang:** Conceptualization, Investigation. **Yanchun Liu:** Writing – review & editing. **Jijun Miao:** Funding acquisition, Resources, Visualization, Writing – original draft, Writing – review & editing.

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Data availability

Data will be made available on request.

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